

Training Deep Neural Networks to Meet Hard Prediction Quotas

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Abstract

A deployed classifier often faces a hard cap on how many items of a known evaluation batch it may assign to a constrained class (a screening pool from which only so many cases can be escalated), optionally with per-group local caps. We call this a *transductive count constraint*. Post-hoc clipping meets the cap but erodes accuracy by editing predictions, while dual-ascent training escalates its multiplier and can stall short of feasibility. We propose **TraLO**, which trains through the count with a bounded penalty, a multiplier that ratchets while the count is violated and freezes once it is met, an optimizer reset at first feasibility, and an undershoot hinge that restores constrained-class recall. Across three MedMNIST datasets and three backbones, TraLO matches the strong constraint-trained baselines over most of the grid and pulls ahead of them on constrained-class F1 where the count binds hard and the class stays discriminable: on OctMNIST tight caps the lead is backbone-general and largest on the ViT-B/16. Grafting the reset and hinge onto the dual baselines transfers most of that lead, while the penalty’s saturating shape proves neutral, so the gain lives in the two portable components, not the formulation the method is named for. Against post-hoc clipping, the constraint-trained family improves macro-F1 on nearly every dataset and backbone while landing on the cap natively. We report the full regime map, wins and ties alike.

1 Introduction

A surprising number of classification systems are deployed under a constraint not on which items get a label but on how many do. A screening program reads a fixed pool of scans and may escalate only so many of them to specialist review, and an audit team can act on only so many flagged filings in a quarter. In each case the operational requirement is a *hard upper bound on the count* of predictions in a constrained class over the evaluation set. The items to be labeled are known in advance (a fixed batch to be scored, not a stream), so the constraint is naturally *transductive*: it is a property of the whole test-set prediction vector, not of any single example.

The standard way to honor such a cap is *post-hoc clipping*: train a normal model, rank the test items by their score for the constrained class, keep the top- K , and reassign the rest. This is trivial and always meets the count, so it is the realistic deployment baseline. But it meets the count by editing predictions, often dozens of them, each a forced error on an item the model was confident about, so accuracy degrades.

The other end of the spectrum is *constrained training*: encode the cap as a constraint and solve the Lagrangian with dual ascent (Fioretto et al. 2020; Hounie, Ribeiro, and Chamon 2023; Chamon and Ribeiro 2020). In principle this is exactly the right formulation, but for a hard count the dual multiplier of a linear penalty climbs without bound: a one-epoch-warmup probe shows it climbing steeply while our bounded multiplier barely moves, for the same final count (Supp. Sec. A). The implementations remain competitive, and we do not over-read the escalation. It is a symptom, not the damage: it buys no faster descent and does not predict where TraLO wins. What it marks is a formulation that on some cells never reaches feasibility at all (Sec. 5.4).

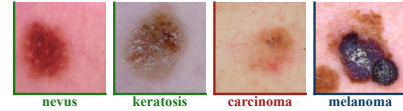


Figure 1: **DermMNIST lesions, benign to malignant** (green frame benign, red malignant, blue the *constrained* class). The task caps how many test images may be labeled melanoma, the minority high-stakes class (11.1% of the test set); TraLO meets that cap *during training* rather than clipping afterward.

We propose **TraLO** (Transductive Loss Optimization), a constrained-training method whose multiplier stays bounded by construction. Its transductive penalty is bounded (a rational-saturation term plus a bounded quadratic in the soft excess over the cap), the multiplier only moves upward while the count is violated and freezes the first time it is met, the Adam state is reset at first satisfaction, and an undershoot hinge restores constrained-class recall once the cap is met. The result is a classifier that lands on the count by itself, with no prediction surgery.

Contributions.

- **A deployable formulation and method.** We cast the operational “ $\leq K$ predictions of class c ” requirement as a transductive count constraint and give a bounded penalty with a freeze-at-satisfaction multiplier, optimizer reset, and undershoot hinge (Sec. 3).
- **A characterized quality win over the trained base-**

lines. Versus the much stronger constraint-trained baselines, TraLO matches constrained-class F1 across most of the grid and beats them where the count binds hard and the constrained class stays discriminable: on all three backbones at the OctMNIST tight caps, with the largest lead on the ViT-B/16. We publish the full regime map, including the large tie region (Secs. 5.1–5.2).

- **A quantified accuracy tax for post-hoc clipping, and a family that avoids it.** The constraint-trained family, TraLO included, improves macro-F1 over the field-default clipping baseline on nearly every dataset and backbone (Sec. 5.3), with native satisfaction on every run in our grid (Supp. Sec. C).
- **Mechanism, ablation, graft.** The bounded penalty reaches the count with a small frozen multiplier; the quality gain is carried by the optimizer reset and hinge, isolated by ablation (Supp. Sec. B) and portable by graft: added to the dual baselines they transfer most of the tight-cap gain, an anti-windup dual restart none (Sec. 5.1).

2 Related Work

Lagrangian and dual constrained learning. Encoding requirements as constraints solved with primal–dual methods is the classical recipe (Bertsekas 1982), brought to deep networks by Lagrangian-duality training (Fioretto et al. 2020) and given statistical footing by PAC constrained-learning theory (Chamon and Ribeiro 2020; Chamon et al. 2022). Resilient Constrained Learning (Hounie, Ribeiro, and Chamon 2023) adapts the constraint level to make hard requirements attainable; PID-style dual updates stabilize the multiplier in safe RL (Stooke, Achiam, and Abbeel 2020); and constraints can beat fixed penalties on sparsity targets (Gallego-Posada et al. 2022). Our critique of runaway escalation is thus specific to the unbounded linear penalty: augmented-Lagrangian and resilient updates are better behaved, yet the resilient method we benchmark still winds up slowly on a hard count (Sec. 5.4). Our two dual-ascent baselines are Fioretto-LDF (Fioretto et al. 2020) and Hounie-RCL (Hounie, Ribeiro, and Chamon 2023); TraLO instead optimizes a bounded penalty and freezes the multiplier at satisfaction.

Output-level and fairness constraints. Closer to our output-level setting, prediction constraints have been enforced during training by primal–dual updates (Nandwani, Pathak, and Singla 2019) and by linear label-count constraints for weakly supervised segmentation (Pathak, Krahenbuhl, and Darrell 2015), or guaranteed per prediction by input-dependent parametrizations (Brosowsky et al. 2021) or an output-layer satisfiability solver (Goyal, Dumancic, and Blockeel 2024). Márquez-Neila, Salzmann, and Fua (2017) find hard enforcement feasible but no better than soft penalties, motivating our bounded penalty. Constrained optimization has protected under-represented classes (Sangalli et al. 2021), and group-wise constraints underpin fair classification (Zafar et al. 2015; Agarwal et al. 2018), which our local caps resemble structurally, though a local cap is a *ceiling*, not a fairness floor (Sec. 4).

Rate-constrained learning. The closest classical relatives constrain prediction *rates*: coverage as a dataset constraint (Goh et al. 2016), non-decomposable rate objectives through surrogate bounds (Eban et al. 2017), non-differentiable rate constraints through proxy-Lagrangian games (Cotter et al. 2019), and generalized rate metrics through multi-player schemes (Narasimhan, Cotter, and Gupta 2019). These bound *expected* rates over a distribution, inductively; ours is a hard count on one specific test set, verified on the argmax predictions and paired with per-group local caps.

Post-hoc correction and resource allocation. Resource-constrained classification allocates a fixed budget over a trained classifier’s outputs at decision time, classically as a small assignment problem, most recently under exactly our local-plus-global structure (Shifman et al. 2025). Related lines fold the budget into learning via adaptive cost-sensitive weights (Shifman et al. 2023) or predict-and-optimize ranking under stochastic capacity (Vanderschueren et al. 2024), and learning-with-rejection (Cortes, DeSalvo, and Mohri 2016) prices abstention per instance, all steering expected costs rather than a hard count. Decision-time allocation always meets the budget by editing predictions; our post-hoc baselines are a greedy threshold heuristic and a linear-programming clipper (Shifman-LP, after Shifman et al. 2025).

Cost-sensitive and imbalanced training. Reweighting or reshaping the loss changes which examples are emphasized: cost-sensitive thresholds (Elkan 2001), focal loss (Lin et al. 2017), class-balanced weights (Cui et al. 2019), label-distribution-aware margins (Cao et al. 2019), prior-corrected logits (Menon et al. 2021). None guarantees the prediction count, which TraLO targets directly.

Transductive and set-level constraints. Transductive inference reasons about a fixed unlabeled test set (Vapnik 1998; Joachims 1999), where soft set-level targets have long existed: expectation regularization pushes predicted class proportions toward a prior (Mann and McCallum 2007), and posterior regularization constrains posterior expectations (Ganchev et al. 2010). Our constraint is transductive in this sense but *hard*: an explicit argmax count budget, with per-group caps, enforced during training.

3 Method

Problem. Let the (known) test set contain N items and let $p_\theta(y = c \mid x_i)$ be the model’s probability for the constrained class c on item i . A *global* count constraint requires the number of items predicted c to satisfy $\sum_{i=1}^N \mathbf{1}[\hat{y}_i = c] \leq K$. We parameterize the cap as a percentage L of the constrained class’s true test-set count, so L measures how much of the class’s actual demand the quota admits ($L30$ admits at most 30% of the true count). *Local* caps also bound the count within each group g : $\sum_{i \in g} \mathbf{1}[\hat{y}_i = c] \leq K_g$ (the same percentage of the group’s own class count). Because the indicator is not differentiable, training uses the *soft count* $S = \sum_i p_\theta(y = c \mid x_i)$, while satisfaction is verified with the *hard count* from $\arg \max$.

A bounded transductive penalty. The defining choice in TraLO is that the constraint penalty is *bounded*. For a soft excess $E = \max(0, S - K)$ over a cap K , the per-constraint penalty is

$$\ell(E; K, \rho) = \underbrace{\frac{E}{E + K}}_{\text{rational saturation}} + \rho \underbrace{\frac{(E/K)^2}{1 + (E/K)^2}}_{\text{bounded quadratic}}, \quad (1)$$

where both terms saturate as $E \rightarrow \infty$ (the first to 1, the second to ρ). What actually drives training is the penalty's *gradient* on the count; differentiating Eq. (1) through $E = \max(0, S - K)$ gives, for $S > K$,

$$\frac{\partial \ell}{\partial S} = \frac{K}{(E + K)^2} + \frac{2\rho}{K} \frac{E/K}{(1 + (E/K)^2)^2}. \quad (2)$$

Both the value and this gradient are uniformly bounded, which keeps the constraint pressure controllable. The total objective combines cross-entropy with the global and local penalties (and, after satisfaction, the undershoot hinge introduced below),

$$L_{\text{total}} = L_{\text{CE}} + \lambda_g \ell_g + \lambda_l \ell_l, \quad (3)$$

where $\ell_g = \ell(E; K, \rho)$ is the global-cap penalty and ℓ_l sums Eq. (1) over the group caps; every cap (the global one and each group cap) carries its own multiplier, so $\lambda_l \ell_l$ abbreviates $\sum_g \lambda_{l,g} \ell(E_g; K_g, \rho)$.

The three statements that follow are elementary: we state them to chart the design space of Eq. (3), synthesizing standard facts rather than claiming new theory.

Lemma 1 (Bounded, saturating pressure) *For $E \geq 0$ and $\rho \geq 0$, the penalty of Eq. (1) obeys $0 \leq \ell < 1 + \rho$, and its count-gradient obeys $0 \leq \partial \ell / \partial S \leq (1 + \rho)/K$, vanishing as $E \rightarrow \infty$. Hence with the multiplier frozen at λ , each cap adds at most $\lambda(1 + \rho)$ to the loss, and its pull on the soft count is at most $\lambda(1 + \rho)/K$: a bounded, saturating pressure independent of the violation size.*

The proof is a short calculus argument (Supp. Sec. A; there the count-gradient peaks in the interior, not at $E=0$, whenever $\rho \gtrsim 0.8$, as throughout our grid). A linear penalty λE instead contributes a gradient constant in E , and dual ascent drives λ upward with no ceiling. The bound also shapes the *updates*, not just the objective:

Proposition 1 (Rank-one constraint gradient) *Each cap's penalty depends on the parameters θ only through its scalar soft-count $S_c(\theta) = \sum_{i \in I_c} p_\theta(y=c | x_i)$. By the chain rule, $\nabla_\theta [\lambda_c \ell(E_c; K_c, \rho)] = a_c \nabla_\theta S_c$ with $a_c = \lambda_c \partial \ell / \partial S_c \geq 0$: at any fixed θ , a nonnegative scalar times the direction $\nabla_\theta S_c$, and that direction is fixed with respect to λ_c, ρ , and the shape of ℓ . The full constraint term therefore lies in the span of $\{\nabla_\theta S_c\}_c$, a subspace of dimension at most the number of caps.*

So the multiplier, ρ , and the shape of ℓ set only the *magnitude* a_c of a cap's push at a given θ , never its direction. This has a sharp consequence once the step is norm-clipped:

Corollary 1 (Clipping erases the multiplier) *Clip the total gradient to max-norm τ . In the regime where one cap's term dominates, $\|a_c \nabla_\theta S_c\| > \tau$ and $\|\nabla_\theta L_{\text{CE}}\|$ is negligible beside it, the clipped step equals $\tau \nabla_\theta S_c / \|\nabla_\theta S_c\|$, independent of λ_c, ρ , and the shape of ℓ (Pascanu, Mikolov, and Bengio 2013).*

In that regime an escalating multiplier cannot change the update: a matched probe with λ differing $\sim 280\times$ leaves training nearly indistinguishable (Supp. Sec. A). The weight ρ and the shape reshape a_c rather than rescale it, so we test them too, and both prove neutral (Sec. 5.1; Supp. Sec. B). Adam's accumulated state and the undershoot hinge are *not* rescalings; they drive a *different* direction, and the quality rests on them.

Multiplier ratchet and freeze. Rather than dual ascent, λ is *ratcheted*: it increases by a fixed step each epoch *only while the hard count violates the cap*, and is *frozen* the moment the count is first satisfied. The bounded penalty means the search ends at a small frozen multiplier rather than an escalating one; multipliers start at 0.05. The quadratic weight ρ ramps linearly (from 5 toward 100) until first satisfaction and then also freezes. Because λ and ρ both stop moving at first satisfaction, TraLO is really a two-phase procedure: while the count is violated it *searches* for a sufficient multiplier by ratcheting λ (the bounded gradient of Eq. (2) keeps that search ending at a small λ), and once the count is met it *settles*.

Stability, and why a verification step is retained. Freezing λ and ρ at first satisfaction turns the moving-target dual problem into a *single fixed* objective $F(\theta) = L_{\text{CE}} + \lambda_g \ell_g + \lambda_l \ell_l$: this is the stability the design buys, the tail of training descending one objective rather than chasing a multiplier that is still moving. That objective is cross-entropy plus a fixed additive penalty of bounded gradient (Lemma 1), smooth away from the caps, so the frozen phase is no worse conditioned than the cross-entropy training the backbone already undergoes. What the penalty is *not* is smooth *at* the cap, and that is what makes feasibility subtle:

Proposition 2 (Kinked penalty, exact at a finite threshold)

The composed penalty $S \mapsto \ell(\max(0, S - K); K, \rho)$ is nonsmooth at the cap: its left-derivative there is 0 and its right-derivative is $1/K$, independent of ρ . Hence along a path on which the soft count rises through K , with cross-entropy preferring a larger count (pull $c = |dL_{\text{CE}}/dS|$ at the cap), the frozen objective $F = L_{\text{CE}} + \lambda \ell$ has a local minimum at $S=K$ once $\lambda \geq cK$: the penalty is of exact-penalty type, feasible at a finite multiplier.

This is the classical exactness of a nonsmooth penalty with nonzero boundary slope (Bertsekas 1982; Nocedal and Wright 2006), and it explains why TraLO reaches the count natively: once cross-entropy saturates (the shared accuracy gate) the pull c collapses, so any frozen $\lambda > 0$ clears the threshold. The design does not lean on this. The ratchet freezes λ at the first *hard*-count satisfaction, which need not reach cK , and the penalty acts on the *soft* count while feasibility is defined by the *hard* count; the verification step (Supp. Sec. C) closes that residual soft-versus-hard gap.

Optimizer reset at satisfaction. At the first epoch the count is met, we reset the Adam optimizer state. This removes the accumulated descent momentum that would otherwise keep pushing the soft count far below the cap (oversatisfying at the cost of constrained-class recall). The reset lets the frozen-multiplier objective settle near the boundary instead of overshooting inward. It is motivated empirically, not derived; the ablation (Supp. Sec. B) isolates its effect.

Undershoot hinge. A satisfied cap can still leave the model under-predicting the constrained class, which costs recall. Concretely, each cap c adds $\lambda_c \beta \max(0, K_c - S_c)/K_c$ to the loss with $\beta=0.5$. It is zero whenever the soft count sits at or above its cap, so it can never re-violate; after satisfaction it gently pushes an undershooting soft count back up toward (but not over) the cap, restoring constrained-class recall. Supp. Sec. B shows the hinge and the reset together carry the effect.

Training schedule. Training has a warmup phase (cross-entropy only, to a stable initialization that is cached and reused) followed by the constraint phase under Eq. (3). A post-hoc verification step is available to enforce a hard count if a run has not landed natively; for TraLO it is essentially never needed (Supp. Sec. C). Supplementary Algorithm S1 summarizes the loop.

4 Experimental Setup

Datasets. We use three MedMNIST v2 (Yang et al. 2023) classification tasks with their official splits (Figure 1 shows DermMNIST examples): **TissueMNIST** (8 classes, constrained class GE, 7.1% of the test set), **DermMNIST** (7 skin-lesion classes from HAM10000 (Tschandl, Rosendahl, and Kittler 2018), constrained class melanoma, 11.1%), and **OctMNIST** (4 retinal-OCT classes, constrained class drusen, 25% of a class-balanced test set). The first two carry the minority class the screening motivation assumes; OctMNIST does not, and we report it because the cap binds hardest there. Local caps use real groups only on DermMNIST (HAM10000 anatomical site: torso/extremity/head-neck); TissueMNIST and OctMNIST lack group metadata and use a fixed synthetic split, a mechanism test rather than a fairness setting (Supp. Sec. E). Caps are set as a percentage L of the constrained class’s true test-set count (Sec. 3); the asymmetric study also varies the group cap G independently of L . The percentage form only standardizes how hard the cap binds across datasets; the method itself receives an exogenous integer K , exactly as a deployed capacity would be given.

Backbones. The headline grid uses three backbones, MobileNetV3 (Howard et al. 2019), RegNetY-400MF (Radosavovic et al. 2020), and a ViT-B/16 transformer (Dosovitskiy et al. 2021), to show the findings are not CNN-specific. All backbones use standard ImageNet-style training conventions (He et al. 2016).

Baselines. Two *post-hoc clippers*: a greedy threshold **Heuristic** and a linear-programming clipper **Shifman-LP** (Shifman et al. 2025). Two *constraint-trained* dual-ascent

methods: **Fioretto-LDF** (Fioretto et al. 2020) and **Hounie-RCL** (Hounie, Ribeiro, and Chamon 2023). We also report **TraLO-bounded**, our own ablation with the undershoot hinge switched off, used both as a method and as the within-cell control in Supp. Sec. B.

Baseline adaptations. The two trained baselines were built for a different setting, so we re-instantiated each on the transductive count task and ran it inside TraLO’s exact harness: the same cached warmup, 300-epoch budget and stop rule (five consecutive feasible epochs), gradient clipping, and post-hoc verification step. Two details are specific to the port: each method’s constraint gradient targets the unlabeled test set, where the count is enforced (the originals target a training or population expectation), and each method’s reported checkpoint is selected on the constraint-excess axis (best feasible, else lowest excess), never on F1, so no baseline is credited with its best-F1 epoch. **Fioretto-LDF** is otherwise unchanged: a linear penalty $\lambda_c S_c$ with per-constraint subgradient ascent $\lambda_c \leftarrow \lambda_c + \eta \max(0, S_c - K_c)$ on the test counts. **Hounie-RCL** keeps its primal/slack/dual updates, with the constraint residual read as $(S_c - K_c)/N$. The one substantive fix in our port is to normalize its primal count-gradient by the sample count so the primal step and the dual update share a scale. Otherwise the primal term is $N \times$ too strong and over-suppresses the capped class. **Shifman-LP** is post-hoc only: a global-and-local linear program that allocates the capped labels on the warmup model’s test-set softmax, using an identity misclassification cost rather than the general cost matrix of the original two-phase allocator (Shifman et al. 2025).

Frozen recipe. All methods share one protocol, fixed before the final runs: ImageNet-pretrained backbones, 224×224 inputs, batch size 64, no augmentation, Adam with learning rate 10^{-4} during warmup and 5×10^{-6} in the constraint phase; warmup = 50 epochs (cross-entropy only), constraint phase = 300 epochs, seeds 1–4 (all reported, no seed selection). Compute is modest: the constraint phase runs a median 2–7 minutes per run on one GPU for TraLO and the dual baselines alike, on top of a shared cached warmup (clipping itself costs seconds but pays the quality tax of Sec. 5.3). Per-method constraint steps were fixed a priori, not tuned per cell: dual step 0.005 (Fioretto-LDF), dual and relaxation steps 0.01 (Hounie-RCL) on all three datasets, and ratchet step 0.002 with hinge weight $\beta=0.5$ (TraLO). A step-fairness sweep (Supp. Sec. A) varies the baseline steps over a $10 \times$ range, and a matching sweep varies TraLO’s own β and λ_{step} (Supp. Sec. B): the OctMNIST tight-cap win is insensitive to both (the best baseline variant still trails by 0.019; TraLO clears the baseline at every arm). The headline grid is *paper_final*: 3 backbones $\times$ 3 datasets $\times$ 9 cap levels $\times$ 6 methods $\times$ 4 seeds = 1944 runs. Models are implemented in PyTorch with torchvision backbones and MedMNIST v2 loaders, and each run trains on a single commodity NVIDIA GPU, with no multi-GPU or distributed training. Code, the configuration generators, and the full per-run results corpus accompany the submission as the Code and Data Supplement.

Metrics. Our *primary* metrics are quality: **constrained-class F1 (cc-F1)**, the quality of the capped class itself and the quantity the constraint most directly puts at risk, and **macro-F1** for overall quality. As *secondary* deployment diagnostics we also report **native satisfaction** (the fraction of runs whose final model meets the global count with no post-hoc edit) and **flips** (the number of test predictions the post-hoc verification step overwrites to enforce the global and per-group caps; 0 when the run needs no adjustment); we treat these as deployment *properties* rather than headline metrics, since every deployable method meets the hard count one way or another (Supp. Sec. C). Gaps versus baselines are *paired* by seed: because the seeds are matched, we summarize a method’s advantage by its seed-winrate and a paired Wilcoxon test rather than by marginal $\pm$ std bars, which would understate a paired effect when a baseline is seed-unstable but its per-seed gap to TraLO is consistent. Claims spanning several cells are tested on cell means (cells are the independent units; the four seeds recur across cells); within one cell, $n=4$ comparisons are reported as effect sizes and seed counts, since a four-pair Wilcoxon cannot fall below $p=0.125$.

5 Results

5.1 Constrained-class quality vs. the trained baselines

We start with constrained-class F1 (cc-F1), the quality of predictions on the very class the cap restricts; Table 1 shows the full symmetric grid, all three datasets over all three backbones, on both headline metrics (cc-F1 here, macro-F1 in Sec. 5.3). TraLO’s sharpest result is where the count binds hard: at OctMNIST tight caps it leads the best constraint-trained baseline on every backbone, by a mean of $+0.038$ across the six cells (sign test $p=0.031$) and peaking at $+0.081$ on the ViT-B/16 (per-backbone paired deltas in Supp. Table S6). Where the cap is slack the summary is a tie: TraLO is best-or-tied against the two trained baselines in 23 of the 27 cells (22 when our own TraLO-bounded ablation variant is also counted as a competitor), and on tissue and derm the differences at equal feasibility are within seed noise (paired Wilcoxon, not significant), a tie that also holds away from the grid, at a tighter ($L40$) and a looser ($L60$) cap on RegNet.

The hard regime: OctMNIST tight caps. The one place TraLO pulls clearly ahead of the trained baselines is OctMNIST at tight caps, where the cap binds hard but the drusen class is still discriminable. There, TraLO leads the best trained baseline on cc-F1 at every backbone and every seed (4/4 per cell, 24 of 24 paired comparisons overall), with the margin largest on the ViT-B/16; the six tight-cap deltas are in Supp. Table S6. Treating cells as the independent unit, the six cell-mean gaps are all positive; the t -test on the cell means gives $p=0.013$, and the same cells improve macro-F1 at five of the six (mean $+0.012$). The win is recall at the cap, not precision: TraLO matches the best trained baseline on constrained-class precision (0.90 vs 0.91) and beats it on recall (0.31 vs 0.27), the quantity the undershoot hinge restores. A tight cap holds recall near its ceiling K/N at high precision, and training recovers the leftover headroom: a property of the cap, not the 28×28 resolution. Figure 2 plots

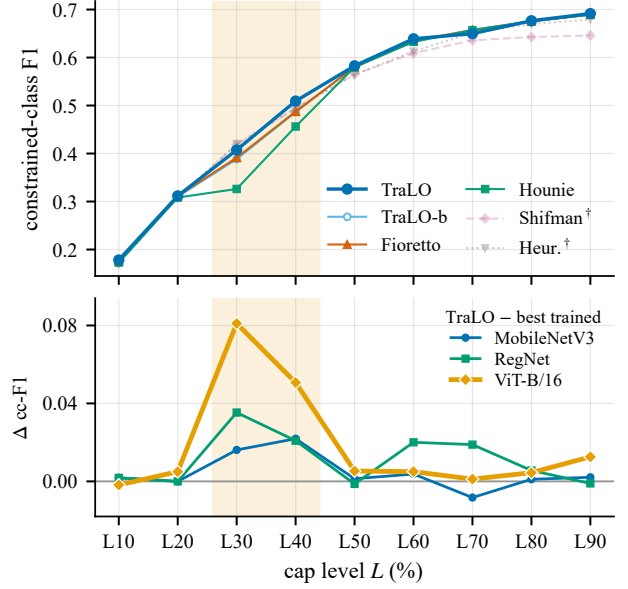


Figure 2: **The OctMNIST tight-cap win is backbone-general and largest on the ViT-B/16.** *Top:* constrained-class F1 of all six methods (MobileNetV3, mean over $n=4$ seeds); the curves nearly coincide, so the win is invisible at this scale (shaded band = tight-binding regime; † faded post-hoc clippers post higher raw cc-F1 by editing). *Bottom:* TraLO minus the best constraint-trained baseline, paired by seed, per backbone; the same inverted-U appears on all three, largest on the ViT-B/16.

the sweep: the six methods nearly coincide (top), while the per-backbone gap (bottom) reveals the win and its inverted-U. The post-hoc clippers’ higher raw cc-F1 in the band is bought by editing predictions (Supp. Sec. C).

Robustness of the OctMNIST win. Three checks. (i) *Backbone-general:* leave-one-backbone-out cross-validation (define the winning band on the two training backbones, test on the held-out one; script in the supplement) holds on every held-out backbone: pooled held-out gap $+0.033$ at a 93% seed-winrate, and 100% on the tight caps themselves. (ii) *Re-derivable without the gap:* a rule blind to the TraLO-vs-baseline gap, “binding (≥ 65 forced flips) with recall headroom at high precision”, selects exactly the winning cells (HIT 100%, MISS 0–40%; label-derived, analysis not deployment). (iii) *Inverted-U, not monotone:* the effect is an inverted-U in tightness, not “tighter is always better,” and it is OctMNIST-specific. This is the one *backbone-general* cc-F1 win; the full regime map follows (Sec. 5.2).

Which components carry the win? A graft experiment. Two controls rerun the six tight-cap cells under the frozen recipe, warmup caches, and one GPU model (144 runs; re-run baselines reproduce the main grid to ± 0.0013 ; per-cell table in Supp. Sec. B). *Graft:* giving each dual baseline exactly TraLO’s optimizer reset and undershoot hinge (its own penalty and multipliers kept) lifts Fioretto-LDF’s cc-F1 in

Table 1: **Quality of the final satisfying predictions on the full symmetric grid:** constrained-class F1 and macro-F1, all three datasets $\times$ all three backbones (mean over seeds 1–4; across-seed stds omitted per cell for width, range .001–.055). **Bold** = best, underline = second-best among the constraint-trained methods; $\dagger$ = post-hoc clippers, for reference; seed-winrates in Supp. Table S6 and Supp. Sec. D. Among trained methods the grid is a tie on Tissue and Derm while TraLO leads every backbone at the OctMNIST tight caps; on macro-F1 TraLO beats the best clipper in 24 of 27 cells, zero losses (mean paired gap +0.031).

Data	Backbone	Cap	Constrained-class F1						Macro-F1					
			Heur. $\dagger$	Shifman $\dagger$	Fioretto	Hounie	TraLO-b	TraLO	Heur. $\dagger$	Shifman $\dagger$	Fioretto	Hounie	TraLO-b	TraLO
Tissue	MNetV3	L30	0.225	0.221	0.275	0.275	0.270	0.275	0.347	0.347	0.405	0.406	0.405	0.408
		L50	0.303	0.300	<u>0.352</u>	0.348	<u>0.352</u>	0.354	0.357	0.357	0.421	<u>0.420</u>	<u>0.420</u>	0.417
		L70	0.344	0.326	0.393	0.392	0.404	<u>0.399</u>	0.362	0.360	0.426	<u>0.427</u>	0.429	0.426
	RegNet	L30	0.241	0.239	0.286	0.291	0.286	0.273	0.329	0.328	0.412	0.410	<u>0.411</u>	0.409
		L50	0.309	0.309	0.354	0.356	<u>0.360</u>	0.366	0.337	0.337	0.421	<u>0.422</u>	0.428	<u>0.422</u>
		L70	0.352	0.349	0.407	0.414	<u>0.412</u>	0.409	0.341	0.340	0.430	0.435	<u>0.432</u>	<u>0.432</u>
	ViT-B/16	L30	0.318	0.320	0.327	<u>0.313</u>	0.327	0.327	0.447	0.447	<u>0.465</u>	0.466	<u>0.465</u>	<u>0.465</u>
		L50	0.411	0.414	<u>0.420</u>	0.422	0.422	0.418	0.458	0.459	0.481	<u>0.482</u>	<u>0.482</u>	0.483
		L70	0.473	0.469	0.466	0.459	<u>0.464</u>	0.462	0.466	0.466	<u>0.487</u>	0.489	<u>0.487</u>	<u>0.487</u>
Derm	MNetV3	L30	0.419	0.417	<u>0.417</u>	0.412	0.419	0.419	0.722	0.721	0.723	0.728	<u>0.727</u>	0.726
		L50	0.557	0.557	<u>0.561</u>	0.560	<u>0.561</u>	0.564	0.747	0.747	<u>0.760</u>	0.761	<u>0.759</u>	0.761
		L70	0.633	0.633	<u>0.645</u>	0.641	<u>0.645</u>	0.648	0.764	0.764	0.784	0.784	0.784	<u>0.781</u>
	RegNet	L30	0.422	0.422	0.419	0.417	<u>0.422</u>	0.424	0.703	0.703	0.705	0.703	<u>0.706</u>	0.708
		L50	0.549	0.548	0.540	0.545	<u>0.539</u>	<u>0.543</u>	0.726	0.726	0.735	0.735	0.735	<u>0.730</u>
		L70	0.617	0.612	<u>0.621</u>	0.620	0.623	0.612	0.740	0.740	0.753	<u>0.752</u>	0.753	<u>0.749</u>
	ViT-B/16	L30	0.371	0.369	0.386	<u>0.384</u>	0.386	0.386	0.646	0.646	0.674	0.674	0.677	<u>0.676</u>
		L50	0.487	0.488	<u>0.507</u>	<u>0.507</u>	0.510	<u>0.507</u>	0.664	0.664	0.700	0.698	0.700	0.700
		L70	0.550	0.542	<u>0.575</u>	0.571	0.578	<u>0.575</u>	0.674	0.672	0.705	0.724	<u>0.707</u>	0.702
OctMNIST	MNetV3	L30	0.420	0.418	<u>0.391</u>	0.326	0.388	0.407	0.666	0.665	<u>0.701</u>	0.678	0.700	0.702
		L50	0.564	0.565	0.580	0.580	<u>0.581</u>	0.583	0.712	0.713	<u>0.762</u>	<u>0.762</u>	<u>0.762</u>	0.763
		L70	0.651	0.636	<u>0.657</u>	<u>0.657</u>	0.658	0.650	0.741	0.737	0.783	0.785	<u>0.784</u>	0.781
	RegNet	L30	0.402	0.400	<u>0.372</u>	0.299	0.369	0.407	0.694	0.693	<u>0.688</u>	0.665	0.686	0.700
		L50	0.536	0.535	<u>0.545</u>	0.541	0.547	0.544	0.734	0.734	<u>0.743</u>	0.742	<u>0.743</u>	0.745
		L70	0.621	0.577	0.624	0.626	<u>0.629</u>	0.645	0.759	0.746	0.768	0.768	<u>0.770</u>	0.777
	ViT-B/16	L30	0.440	0.437	0.335	0.332	<u>0.352</u>	0.422	0.718	0.717	0.692	0.691	<u>0.697</u>	0.722
		L50	0.600	0.600	0.592	0.588	<u>0.595</u>	0.597	0.771	0.771	<u>0.779</u>	0.776	<u>0.779</u>	0.781
		L70	0.671	0.653	<u>0.671</u>	0.670	<u>0.671</u>	0.672	0.795	0.790	0.805	<u>0.804</u>	0.805	0.805

six of six cells (+0.019 to +0.073) and Hounie-RCL’s in five of six. Measured seed-by-seed against the stronger of the two grafted baselines, TraLO’s edge is only +0.0015 and it trails in two of the six cells: once the recipe is grafted the constraint-trained methods are essentially interchangeable, which is exactly our claim: the reset and hinge, not TraLO’s formulation, carry the gain. *Anti-windup*: Fioretto-LDF with dual restarts (Gallego-Posada et al. 2022) deploys predictions identical to plain Fioretto-LDF in every cell (the deployed checkpoint is fixed before first feasibility), and TraLO beats it in all six, so the deficit is not missing anti-windup machinery. *Penalty shape*: swapping Eq. (1)’s saturating penalty for a linear one at the same ratcheted multiplier moves cc-F1 by -0.001 (the bounded shape wins 9 of 24 runs and loses 9, with 6 exact ties; sign-test $p=1.0$) and feasibility by two epochs, consistent with the fixed-direction argument of Sec. 3. The quality mechanism is the reset plus hinge, portable across hosts; Eq. (1) buys a small, readable multi-

plier, not the gains.

5.2 Wins, ties, and losses across the full grid

Across the full corpus the pattern is consistent: **TraLO wins where the count binds hard and the constrained class stays discriminable, and ties where either condition fails.** Binding alone is not enough: at $L30$ the warmup model over-demands the cap by $2.4\text{--}4.8\times$ on tissue and derm, more than OctMNIST’s $2.0\text{--}2.5\times$, yet only OctMNIST clears the win bar. We score the *paper_final* grid on the three backbones plus dedicated MobileNetV3 asymmetric sweeps, partitioned into five constraint-geometry regimes, every regime reported (per-cell breakdown in Supp. Sec. D). A regime counts as a *win* only when the mean paired cc-F1 gap and at least half its cells clear +0.005, and a *loss* under the mirror-image rule; no regime mean falls below -0.005 . Only two regimes clear the win bar: OctMNIST tight-symmetric caps (the backbone-general win of Sec. 5.1, largest on ViT-B/16)

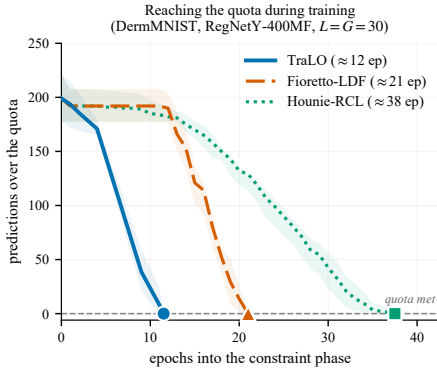


Figure 3: **TraLO reaches the count fast and holds it.** Best-so-far constraint excess (lowest predictions over the cap so far) over the constraint phase; DermMNIST (RegNetY-400MF), tight cap $L=G=30$. Median over $n=4$ seeds; band: interquartile range; markers at each method’s median epoch-to-feasibility. All three start about 200 over the cap; on this cell TraLO’s ratchet reaches the cap first, the zero-start duals later. Giving the baselines TraLO’s nonzero start narrows but does not close this (Supp. Sec. B).

and the global-tighter asymmetric ($G < L$) caps of DermMNIST (+0.8pp on MobileNetV3, reproduced on RegNet and tied on ViT; *preliminary*, as only MobileNetV3 has the full asymmetric grid). Every other regime, including all symmetric caps, is a statistical tie (symmetric tissue+derm mean cc-F1 gap -0.0013 , seed winrate 29%, $n=216$), and none is a loss. Against the post-hoc clippers macro-F1 is higher for TraLO on every regime (Sec. 5.3).

5.3 Quality vs. post-hoc clipping

The other baseline family is the field default: train unconstrained, clip the top-scoring predictions down to the cap. Both end up meeting the count; the question is what quality that costs. Clipping meets the cap by overwriting predictions (often dozens per run), an accuracy tax that TraLO, landing on the count natively, avoids. Across the full 1944-run grid (run-level, paired by seed, against the best post-hoc clipper) this is a macro-F1 gain for TraLO of +0.053 (Tissue), +0.016 (Derm) and +0.021 (OctMNIST), positive in every regime and in 77 of the 81 symmetric cells (per cell at the displayed caps in the macro-F1 block of Table 1; Supp. Sec. D has all 81; the exceptions are the very tightest $L10/L20$ caps), with constrained-class F1 moving the same way in aggregate. The margin over clipping is a property of training under the constraint rather than of TraLO alone: the dual baselines post nearly identical macro-F1 margins (the lone exception a tie at OctMNIST×ViT), so we claim it for the constraint-trained family; constraint-time training also adapts to the test batch, a confound we do not separate from avoided editing. What is TraLO’s own is reaching that quality with the fastest and most reliable native convergence (Sec. 5.4) and the fewest prediction edits at binding caps. The tax is steepest where the cap binds: at the tight $L30$ cap, clipping costs up to 0.061 macro-F1 (Tissue; Supp. Table S3). How TraLO

avoids the tax (near-perfect native satisfaction, an order of magnitude fewer edits) is a deployment property reported in Supp. Sec. C: what separates the methods is the quality each preserves while meeting the count.

5.4 Convergence

These advantages rest on TraLO reaching the count during training. Figure 3 tracks how far each method’s prediction count sits above the cap over the constraint phase, on the cell where the three methods separate most cleanly: DermMNIST on RegNet, whose cross-entropy saturates slowly enough that the count and the task objective openly compete. TraLO’s ratcheted multiplier acts from the first constraint epoch and reaches the cap well before either zero-start dual baseline (per-method epochs in the caption). The gap holds grid-wide: native feasibility is 100% for TraLO and Hounie-RCL but only 87% for Fioretto-LDF, whose unbounded multiplier stalls short of the cap on the rest. Across DermMNIST and TissueMNIST on the two convolutional backbones, over the 18 symmetric-cap cells we log, Hounie-RCL is the slowest of the three in every one and TraLO is never the slowest: TraLO reaches the count first or within one epoch in 15 of the 18 (strictly first in 11), at median epochs-to-feasibility of 17, 20, and 49.5 for TraLO, Fioretto-LDF, and Hounie-RCL. Fioretto-LDF is competitive when it converges but stalls on the runs it does not, so TraLO is the only method that reaches the count both fast and on every run, which lets it avoid the editing that costs clipping its quality. The per-cell census accompanies the released corpus.

6 Conclusion

Count caps are a real, common deployment constraint whose two standard answers each cost something: post-hoc clipping pays an accuracy tax for editing predictions, and dual-ascent training chases the cap with an escalating multiplier and, in its linear-penalty form, can stall short of it. TraLO trains through the count with a bounded penalty, a multiplier frozen at satisfaction, an optimizer reset, and an undershoot hinge. It matches the strong constraint-trained baselines across the grid and beats them on the constrained class where the cap binds hard and the class stays discriminable (OctMNIST tight caps; preliminarily, DermMNIST’s global-tighter asymmetric caps), and improves macro-F1 over post-hoc clipping while landing on the count natively. An ablation and a cross-method graft trace that gain to the reset and hinge, which transfer to the dual baselines; the penalty’s own shape proves neutral (Sec. 5.1).

Limitations and broader impact. The advantage over trained baselines is regime-specific, and the DermMNIST asymmetric win is MobileNetV3-only. All three datasets are MedMNIST-derived (28×28 upsampled to 224); full-resolution validation and a theoretical ranking against other solvers remain future work. TraLO is transductive, scoped to a batch scored in one shot, not a stream. A cap trades recall for capacity, and our local caps are *ceilings*, not *fairness floors* (Hardt, Price, and Srebro 2016).

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